WASP-50b

R-1541

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-50_150112` · created 2026-08-01T19:10:38+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457034.721 +/- 0.072 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.051
Transit depth (Rp/Rs)^2	2.43 +/- 1.59 [%]
Orbital Inclination (inc)	86.3 +/- 2.82
Airmass coefficient 1 (a1)	0.92 +/- 0.23
Airmass coefficient 2 (a2)	0.057 +/- 0.078
Scatter in the residuals of the lightcurve fit is	26.29 %
Best Comparison Star	#3 - [168, 237]
Optimal Aperture	4.37
Optimal Annulus	8.21
Transit Duration (day)	0.052 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.156 ± 0.051 vs 0.1404 (deviation 0.29σ)
Duration fit vs published	0.052 d vs 0.0754 d (deviation 0.68σ)
Mid-time O–C	19.0 min
Depth SNR	2.9
Residual scatter	26.29 %

Light curve

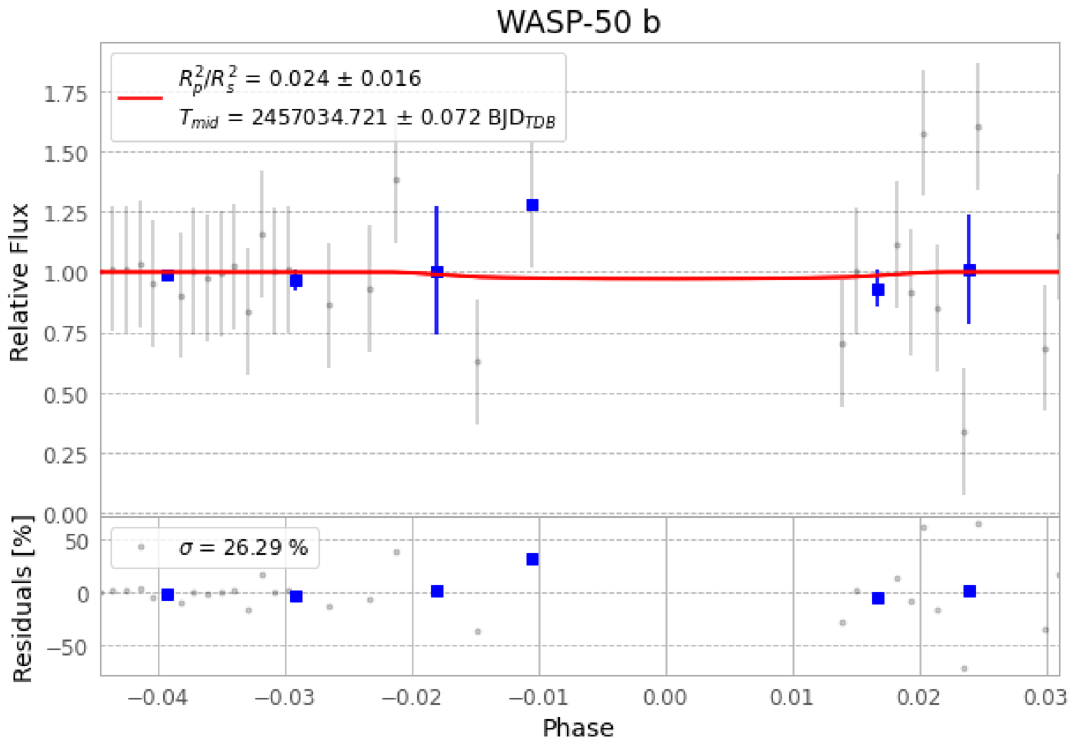

Figure 1 — FinalLightCurve_WASP-50 b_2015-01-11.png (SHA-256 6392cad9664eab8a…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [225, 299], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 1e6b38c1ad905b67…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

77 FITS frames (51 MB), `WASP-50150112030912.FITS` → `WASP-50150112065712.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-150112.log (cb8d0996d961…), FinalLightCurve_WASP-50 b_2015-01-11.png (6392cad9664e…), FinalLightCurve_WASP-50 b_2015-01-11.csv (dea70606fb01…)

Compute resources

Wall time 235.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)